An ABC Rejection Algorithm for Signature Matching

This is my understanding of an Approximate Bayesian Computing (ABC) algorithm that aims to find the mean injection level a for a mutational signature arising from

$$\text{signature} = (1-\alpha)(0.4\text{SBS}_1 + 0.4\text{SBS}_5 + 0.2\text{SBS}_{18}) + \alpha\text{SBS}_{88}, \tag{4}$$

where $\alpha \sim \mathcal{N}(a, a/10)$ with support $\alpha \in [0,1]$. The parameter α can be drawn from any appropriate distribution, e.g. lognormal or beta.

1. Why ABC? ABC methods only need access to a data generator - i.e. the ability simulate observations. Unlike typical Bayesian inference, it does not need access to the likelihood function[10].

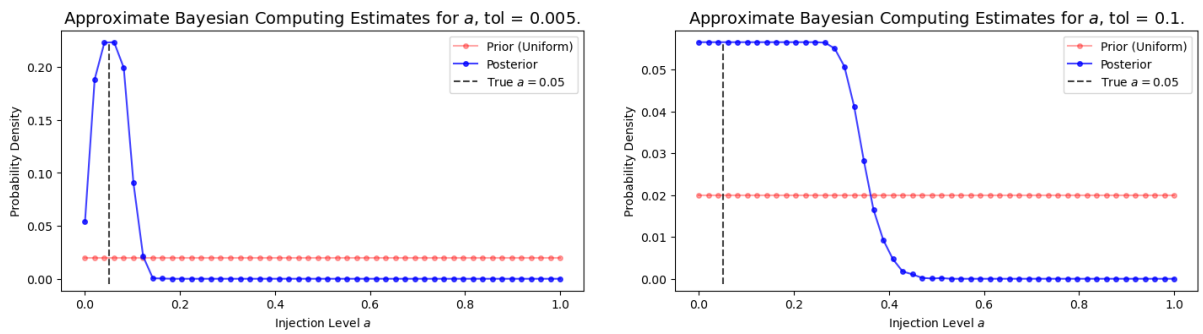
We have the ability to (easily) generate (lots of) synthetic mutational signatures for different values of a.

2. Methodology. Let V denote an observed mutational signature generated from (4) with an unknown a. Assume a uniform prior for a on $[0,1]$ - that is, all values of a are equally likely. Then, for each a,

1. Generate K synthetic mutational signatures $(V_a^j)_{j=1}^K$ with injection $\alpha \sim \mathcal{N}(a, a/10)$.
2. Compare each generated signature to V via cosine similarity, i.e. find $\cos(V, V_a^j)$.
3. If $\cos(V, V_a^j) < \varepsilon$, where ε is some tolerance (e.g. 0.005), *accept* this a-value (the distributions are similar). Otherwise, *reject* this a-value.
4. Count the number of acceptances for each a. Scale to produce an approximate distribution for a, as in Figure 16.

In the notebook, $K = 1000$, there are 250 values of a evenly spaced in $[0,1]$, and the tolerance is set at $\varepsilon = 0.005$.

The choice of tolerance is important. If it's too large, the estimate for a becomes wide and uninformative. If it's too small, the algorithm may not accept any proposals.



(a) Reasonable tolerance.

(b) Too large tolerance.

Figure 16: ABC estimates for a under different tolerance levels.

See Figure 17 and Figure 18 for two more examples.

[10] Normally, posterior $\propto$ likelihood·prior. Loosely speaking, the simulated observations can be thought of as being drawn from the likelihood function, eliminating the need for direct access to the likelihood.

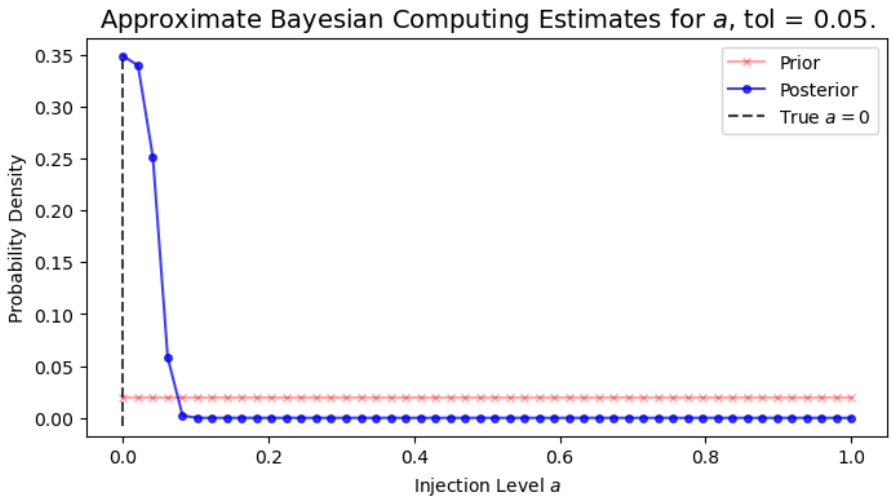


Figure 17: ABC estimates for the injection level (true $\alpha = 0$, tolerance 0.05).

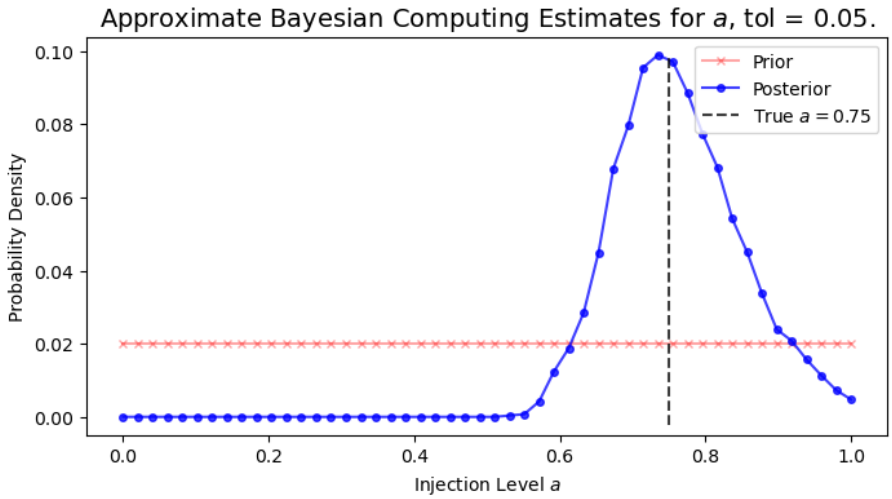


Figure 18: ABC estimates for the injection level (true $\alpha = 0.75$, tolerance 0.05). Here you could make an argument that the tolerance is not small enough.

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